

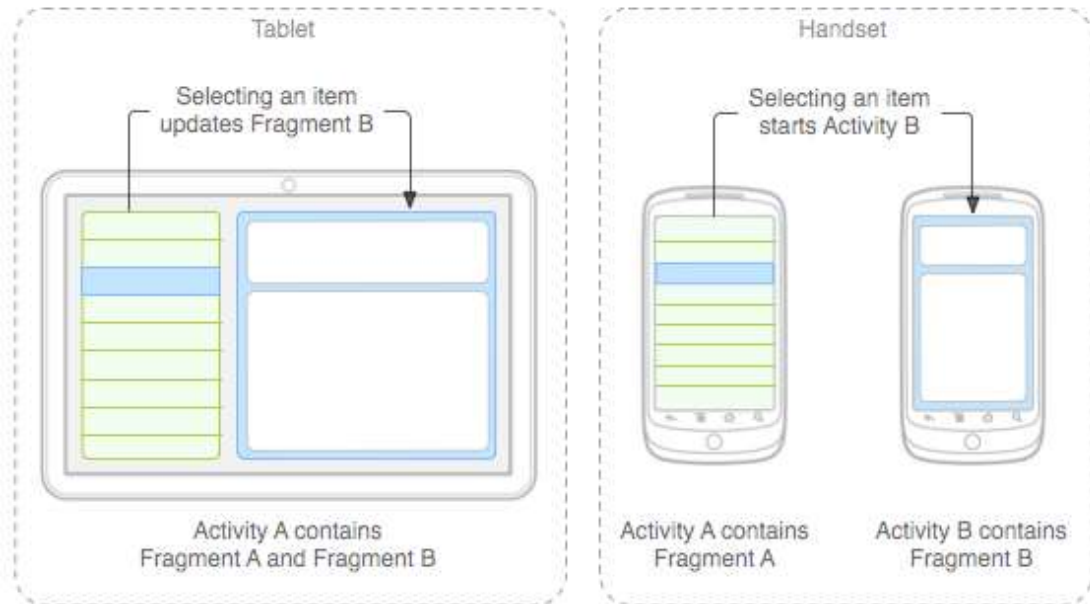
Fragment

Contents

- What is Fragment
- Create Fragment
- Interact with Activity

Fragment

- **Fragment: A reusable segment of Android UI that can appear in an activity.**
 - can help handle different devices and screen sizes
 - can reuse a common fragment across multiple activities



Create Fragment

```
public class Name extends Fragment {  
    @Override  
    public View onCreateView(LayoutInflater inflater,  
        ViewGroup vg, Bundle bundle) {  
        // load the GUI layout from the XML  
        return inflater.inflate(R.layout.id, vg, false);  
    }  
  
    public void onActivityCreated(Bundle savedInstanceState) {  
        super.onActivityCreated(savedInstanceState);  
        // ... any other GUI initialization needed  
    }  
  
    // any other code (e.g. event-handling)  
}
```

Add Fragment to Activity

- **Static way:**

```
<LinearLayout
```

```
    android:layout_width="match_parent"
```

```
    android:layout_height="match_parent">
```

```
        <fragment android:name="com.example.news.ArticleListFragment"
```

```
            android:layout_width="match_parent"
```

```
            android:layout_height="match_parent" />
```

```
</LinearLayout>
```

Add Fragment to Activity

- **Dynamic way:**

```
FragmentManager fragmentManager = getFragmentManager();  
FragmentTransaction fragmentTransaction =  
fragmentManager.beginTransaction();
```

```
ExampleFragment fragment = new ExampleFragment();  
fragmentTransaction.add(R.id.fragment_container, fragment);  
transaction.addToBackStack(null);  
fragmentTransaction.commit();
```

Communicate with Activity

- **From Activity to Fragment:**

- keep fragment instance in Activity and call fragment methods directly

- **From Fragment to Activity**

- Pass a callback when the Fragment is created

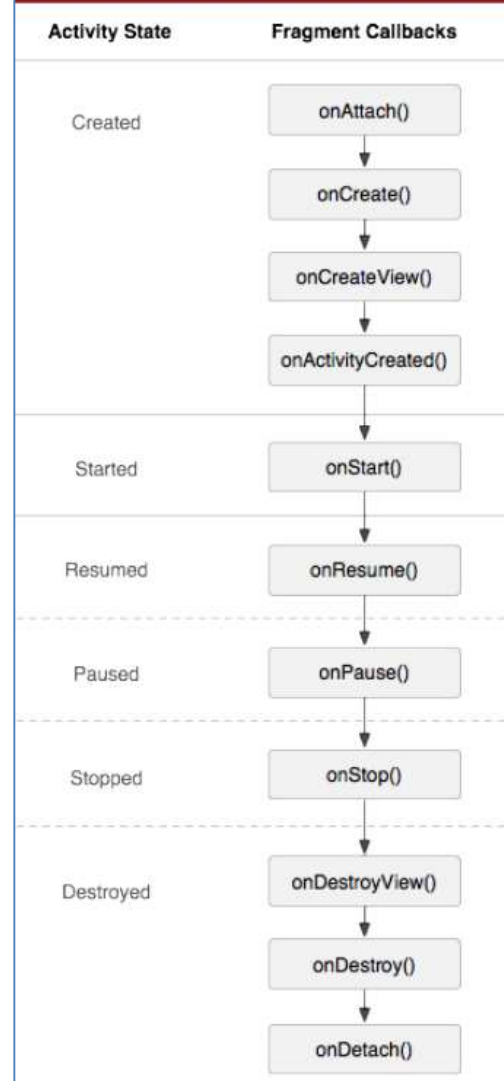
```
public class HeadlinesFragment extends Fragment {
    OnHeadlineListener callback;
    public void setOnHeadlineListener(OnHeadlineListener callback)
    {
        this.callback = callback;
    }
    public interface OnHeadlineListener {
        public void onArticleSelected(int position);
    }
}
```

```
public static class MainActivity extends Activity
implements OnHeadlineListener {
    @Override
    public void onAttachFragment(Fragment fragment) {
        if (fragment instanceof HeadlinesFragment) {
            HeadlinesFragment headlinesFragment =
                (HeadlinesFragment) fragment;

            headlinesFragment.setOnHeadlineListener(this);
        }
    }
}
```

Fragment lifecycle

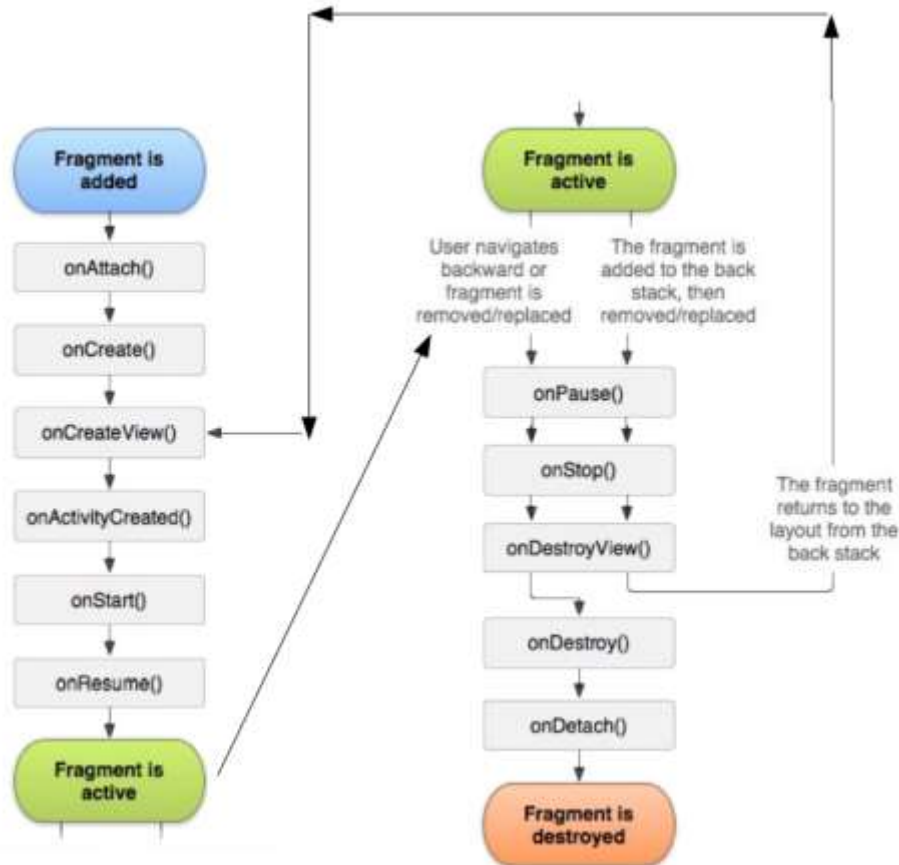
- Fragments have a similar life cycle and events as activities.
- Important methods:
 - onAttach to glue fragment to its surrounding activity
 - onCreate when fragment is loading
 - onCreateView method that must return fragment's root UI view
 - onActivityCreated method that indicates the enclosing activity is ready
 - onPause when fragment is being left/exited
 - onDetach just as fragment is being deleted



Fragment lifecycle

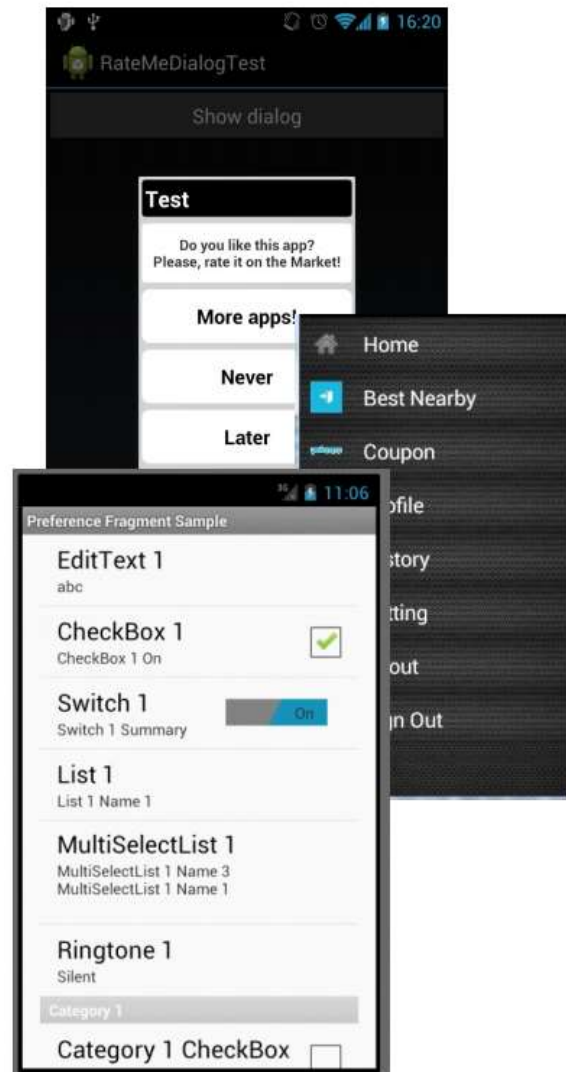
- **Fragment in an Activity---Activity Lifecycle influences**
 - Activity paused -> all its fragments paused
 - Activity destroyed -> all its fragments destroyed
 - Activity running -> manipulate each fragment independently.
- **Fragment transaction: add, remove, etc.**
 - adds it to a back stack that's managed by the activity—each back stack entry in the activity is a record of the fragment transaction that occurred.
 - The back stack allows the user to reverse a fragment transaction (navigate backwards), by pressing the Back button

Fragment lifecycle



Fragment subclasses

- **DialogFragment** - a fragment meant to be shown as a dialog box that pops up on top of the current activity.
- **ListFragment** - a fragment that shows a list of items as its main content.
- **PreferenceFragment** - a fragment whose main content is meant to allow the user to change settings for the app



Classwork

- **Using Fragment to make:**
 - A ListProduct Fragment
 - A DetailedProduct Fragment
- **Click an Item in ListProject Fragment, show DetailedProduct Fragment**

References

- <https://google-developer-training.github.io/android-developer-fundamentals-course-concepts-v2/>